Longitudinally Guided Post-Contrast MRI Synthesis via a Schrödinger Bridge

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Impact

Synthesizing post-contrast MRI from pre-contrast imagery could allow patients to avoid repeated gadolinium injections, but enhancement is not always evident in pre-contrast images. Conditioning on a prior scan supplies additional enhancement information that improves synthesis quality.

Synopsis

Motivation: Synthesizing post-contrast enhanced images could avoid gadolinium injection, but enhancement is not always predictable from pre-contrast information. Many patients who are under surveillance have a prior contrast-enhanced exam that encodes enhancement for similar anatomy.
Goal(s): Exploit prior exam information in a contrast synthesis task.
Approach: The Image-to-Image Schrödinger Bridge (I^2SB) algorithm is used to map current T1 to contrast-enhanced T1 (CT1) images conditioned on FLAIR/T1/T2 imagery. To extend this to the longitudinal synthesis task, five adjacent T1/CT1 slices are provided as side information during reconstruction.
Results: Longitudinal I^2SB produces higher quality CT1 images while remaining robust towards uninformative guide information.

Introduction

Brain metastases are monitored with serial gadolinium-enhanced MRI[1], and reducing repeated gadolinium exposure[2] for patient comfort (and reduced visit-time) motivates synthesizing post-contrast images from non-contrast ones. Single-exam synthesis must infer enhancement from T1, T2 and FLAIR alone. In surveillance, however, every patient has a prior post-contrast exam, highly correlated with the current one. We condition an image-to-image Schrödinger bridge (I^2SB)[3] on that prior exam and evaluate it under inference-realistic conditions.

Methods

Image-to-Image Schrödinger Bridge: I^2SB seeks to take some image $\boldsymbol{x}_1$, sampled from some known distribution, and estimate some related image $\boldsymbol{x}_0$ coming from some alternative target distribution. Generating a sample $\boldsymbol{x}_t$ from the bridge involves generating noisy convex combinations of $\boldsymbol{x}_0$ and $\boldsymbol{x}_1$, where the additive noise power is time varying for $t \in [0, 1]$. A network is trained to, given bridge sample $\boldsymbol{x}_t$, predict the bridge endpoint $\boldsymbol{x}_0$. The algorithm itself iteratively predicts $\boldsymbol{x}_0$ and then resamples a new bridge point $\boldsymbol{x}_t$ while annealing the time index to 0 (Figure 1). In our case, we bridge from the current pre-contrast image to the desired post-contrast image. The original I^2SB paper[3] uses a UNet[4] with an appended time-dependent channel. Since we have access to other contrasts of the same anatomy, we additionally append [T1, T2, FLAIR] as input channels, so the model need only learn the enhancement pattern from T1 to CT1.

Longitudinal Conditioning: Where enhancement is not evident in the pre-contrast images, the patient's prior contrast-enhanced exam can supply it. The bridge endpoints are unchanged, and the prior exam enters only as conditioning: its T1 and CT1 at a 5-slice window are appended as ten further input channels. Including the neighboring slices makes the network robust to misregistration between slices (through-plane misregistration).

Experiments: We trained on NYUMets[5], a longitudinal dataset of brain-metastasis patients with FLAIR, T1, CT1 and T2 at each visit, split into a training set of 2318 exams and a holdout set of 1007 exams. Within each exam the contrasts are already co-registered and skull-stripped. We train two regressors over the same training set while only giving one access to prior-exam data. At inference, the prior exam is registered to the current one by a 3-D affine fit learned only from pre-contrast images, without relying on the CT1 being predicted. PSNR, SSIM and NRMSE were computed within the brain mask and averaged per exam, and paired Wilcoxon signed-rank tests were run over the holdout set.

Results

Across 1007 test exams, conditioning on the prior exam improved every metric (Figure 3): PSNR rose from 24.79 ± 0.09 to 25.73 ± 0.10 dB, SSIM from $87.79 \pm 0.21\%$ to $90.83 \pm 0.19\%$, and NRMSE fell from $13.07 \pm 0.19\%$ to $11.97 \pm 0.19\%$ (mean $\pm$ s.e. over exams; all $p < 10^{-3}$, paired Wilcoxon signed-rank test). The longitudinal model was better in 97% of exams (median PSNR gain +0.84 dB). Appending input channels to the UNet comes at little computational cost, so both methods are effectively computationally matched.

Figure 2 shows visual improvement. In the first row, the single-study I^{2}SB result overpredicts the enhancement details whereas longitudinal I^{2}SB better preserves anatomical detail. The second row gives an example where a lesion appears in the CT1 image and not in the current T1. Single-study I^{2}SB predicts the lesion, but longitudinal I^{2}SB recovers its shape more accurately. Additionally, the prior CT1 guide seems to contain enhancement details near the brain stem that do not appear in the ground truth; longitudinal I^{2}SB is robust to them. In the third row, single-study I^{2}SB hallucinates a lesion absent from the acquired CT1, which the guide corrects.

Figure 4 shows cases where the guide contains irrelevant or incorrect enhancement. Longitudinal I^{2}SB remains visually comparable to single-study I^{2}SB, demonstrating robustness to guide quality. In some cases, longitudinal I^{2}SB performs slightly better, likely due to the guide still encoding structural details that may be absent from the pre-contrast input. Figure 5 quantifies this by scoring the registered prior CT1 against the current CT1. The gain grows from $\approx +0.3$ dB for the least similar priors to +2.5 dB for the most similar, and stays positive on average in every bin.

Discussion

The prior exam is routinely available in surveillance, yet it is generally unused by contrast-synthesis methods. As conditioning, it lets the network transfer known enhancement rather than infer it, and a poor prior reduces the benefit without degrading synthesis on average. However, registration quality here is generally high, and performance may degrade in anatomies less rigid than the brain.

Conclusion

Longitudinally guided I^{2}SB improves synthesis of follow-up post-contrast T1 images by leveraging patient-specific information from prior examinations, with inter-exam registration based solely on non-contrast images.

Acknowledgements

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References

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Figures

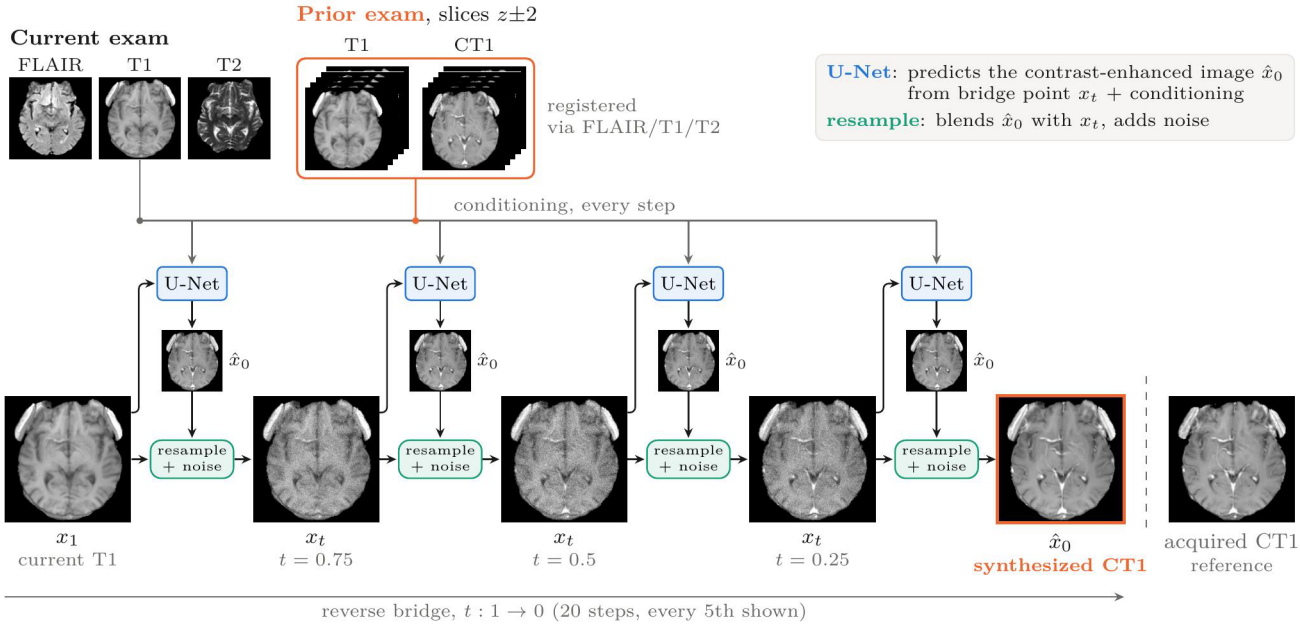


Figure 1: Longitudinal I^{2}SB (L-I^{2}SB) synthesis of the post-contrast T1 (CT1). The reverse bridge runs from the acquired T1 (x_1, t=1) to the synthesized CT1 ($\hat{x}_0$, t=0) in 20 steps; four states are shown. At each step the UNet predicts $\hat{x}_0$ from x_t, the current FLAIR/T1/T2, and the registered prior T1 and CT1 (slices $z-2 \ldots z+2$). Images are one test exam's trajectory; the acquired CT1 (right) is for reference only.

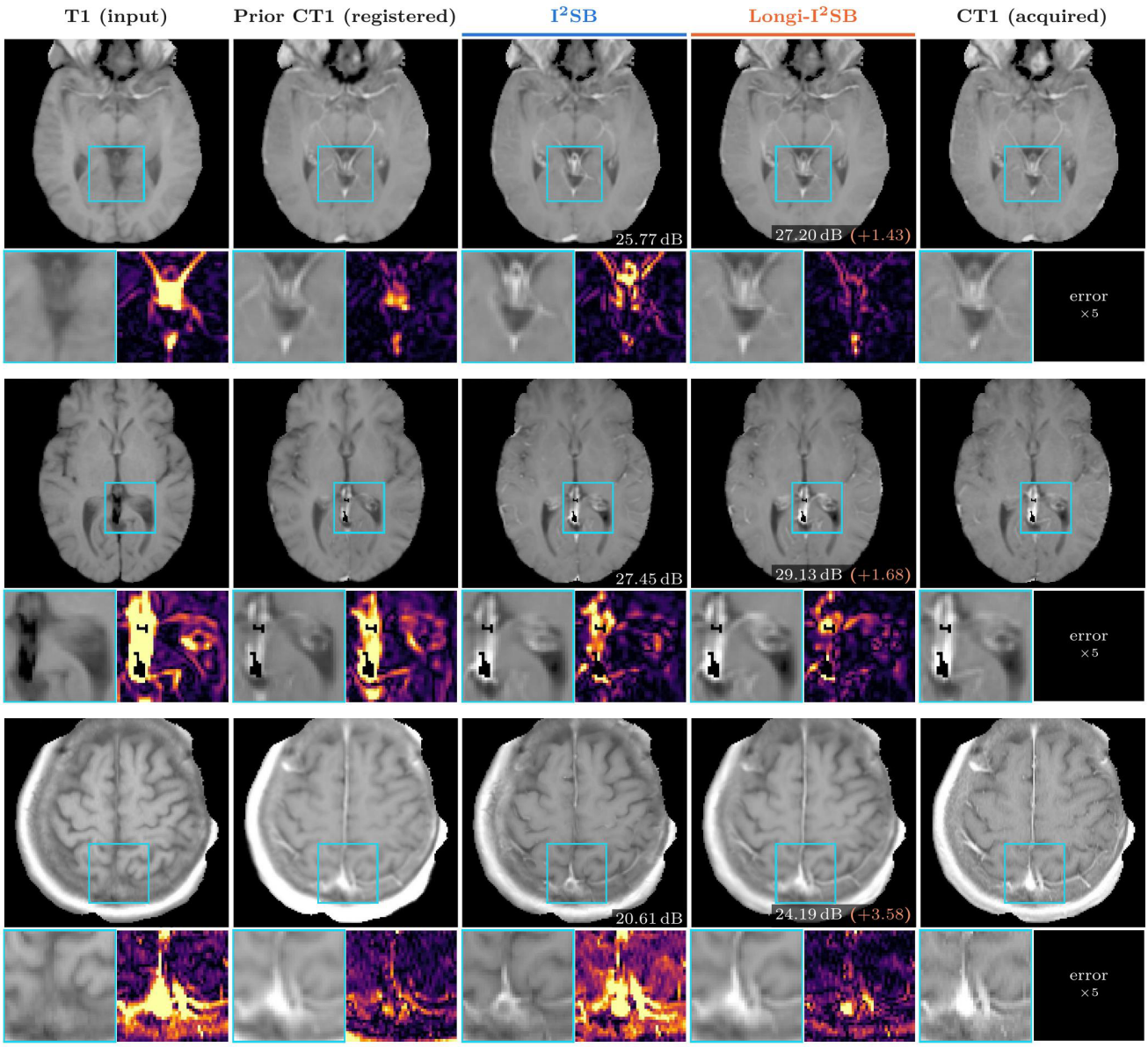


Figure 2: Example test cases (rows). Columns: current T1 (input), registered prior CT1, I^{2}SB without the prior exam, L-I^{2}SB (longitudinal I^{2}SB, with it), and the acquired CT1 (truth). Beneath each image: the magnified green box and its 5× residual. PSNR (dB) is printed on each synthesized image.

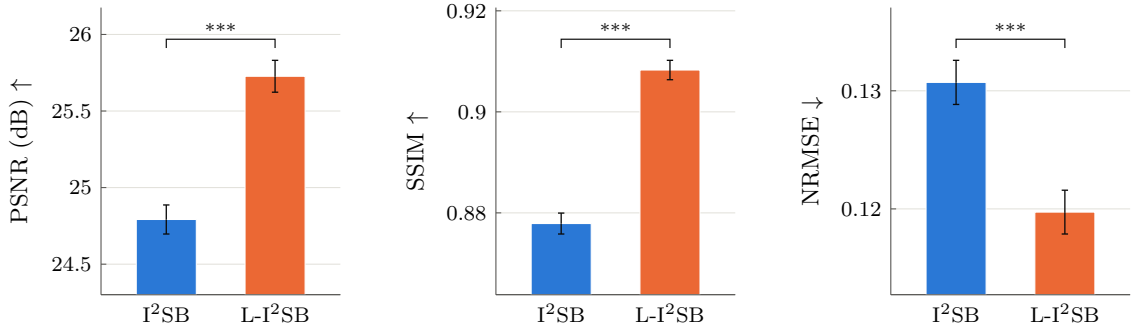


Figure 3: 1007 test studies, slices 70–109, metrics within the mask; L-I^{2}SB is longitudinal I^{2}SB. Mean ± s.e. over studies; *** paired Wilcoxon signed-rank $p \leq 0.001$ (Holm-adjusted); value axes do not start at zero.

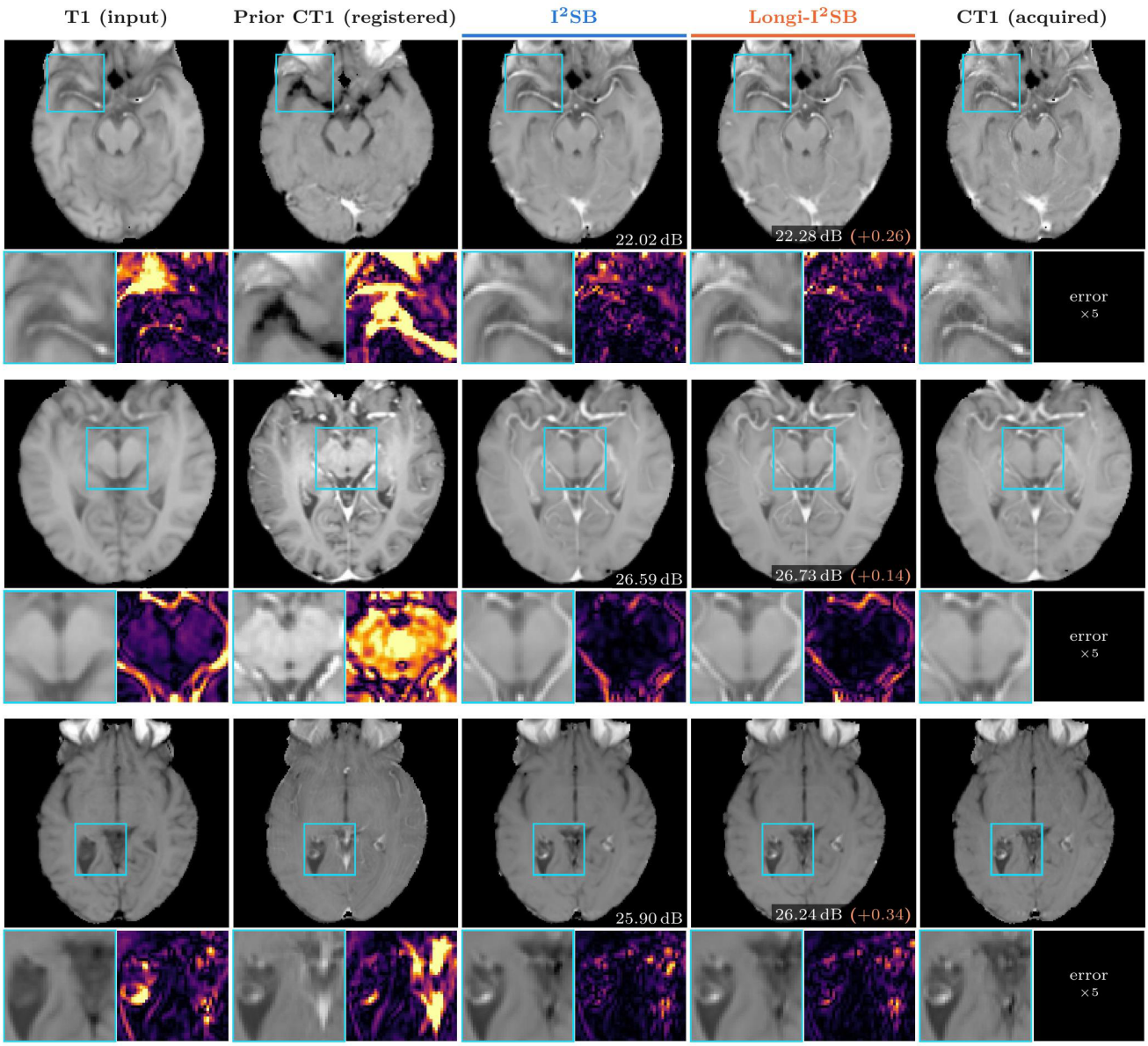


Figure 4: Robustness to a poor prior exam: cases where the registered prior CT1 differs greatly from the current one. Columns and panels as in Figure 2; L-I^{2}SB is longitudinal I^{2}SB.

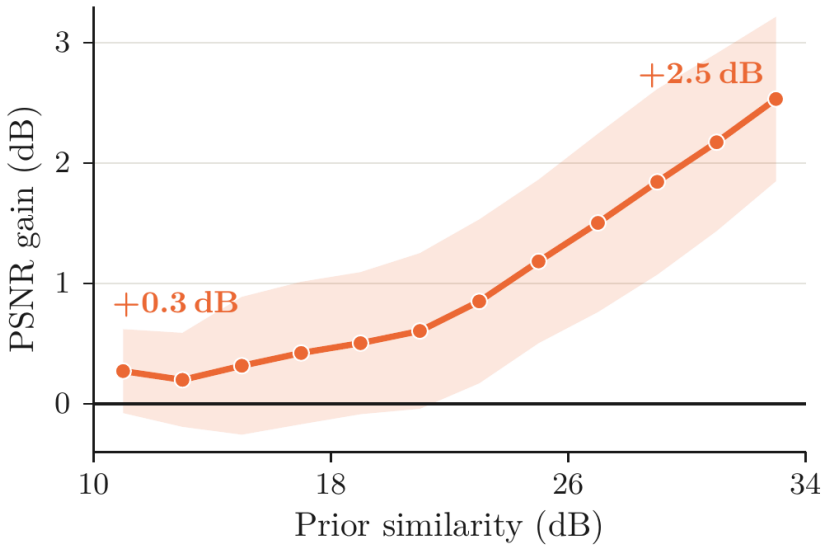


Figure 5: PSNR gain of L-I^{2}SB (longitudinal I^{2}SB) over single-exam I^{2}SB versus prior similarity, the PSNR of the registered prior CT1 against the current CT1 (same mask and data range as the reported metrics). Line: mean over slices in 2 dB bins; band: $\pm$ 1 s.d. The gain grows with prior similarity, from +0.3 dB to +2.5 dB, and is positive on average in every bin.